Investigate the functioning of a rootkit and demonstrate techniques to detect it

- **What is a Rootkit?**
  - A **rootkit** is a type of **malicious software** designed to:
  - Gain **unauthorized root/admin access**
  - **Hide its presence** from users and security tools
  - Maintain **persistent control** over a system
  - It operates stealthily, making detection difficult.
- Types of Rootkits
  - **User-mode rootkit** – runs like normal applications
  - **Kernel-mode rootkit** – modifies OS kernel (very dangerous)
  - **Bootkit** – infects bootloader
  - **Firmware rootkit** – infects BIOS/UEFI

Step-by-Step Working:

- **Initial Infection**
  - Via phishing, malicious downloads, exploits
- **Privilege Escalation**
  - Gains root/admin access
- **Persistence**
  - Installs itself in system files or kernel
- **Stealth Mechanism**
  - Hides files, processes, and network activity
  - Hooks system calls
- **Backdoor Creation**
  - Allows attacker remote access anytime

DEMONSTRATION of ROOTKIT SCANNER TOOLS

- Install Rootkit Scanner Tools
  - sudo apt update
  - sudo apt install chkrootkit rkhunter -y
- Detection Techniques
- Using chkrootkit : sudo chkrootkit
- Output shows:
  - INFECTED / suspicious entries
  - Hidden files or processes
- Using rkhunter: sudo rkhunter --check
- Update database: sudo rkhunter –update
- Detects:
  - File integrity issues
  - Suspicious kernel modules
- **Check Network Connections**
  - netstat -tulnp
  - Look for:Unknown ports, Suspicious connections
- File Integrity Check

- sudo debsums -s
- Detects modified system files